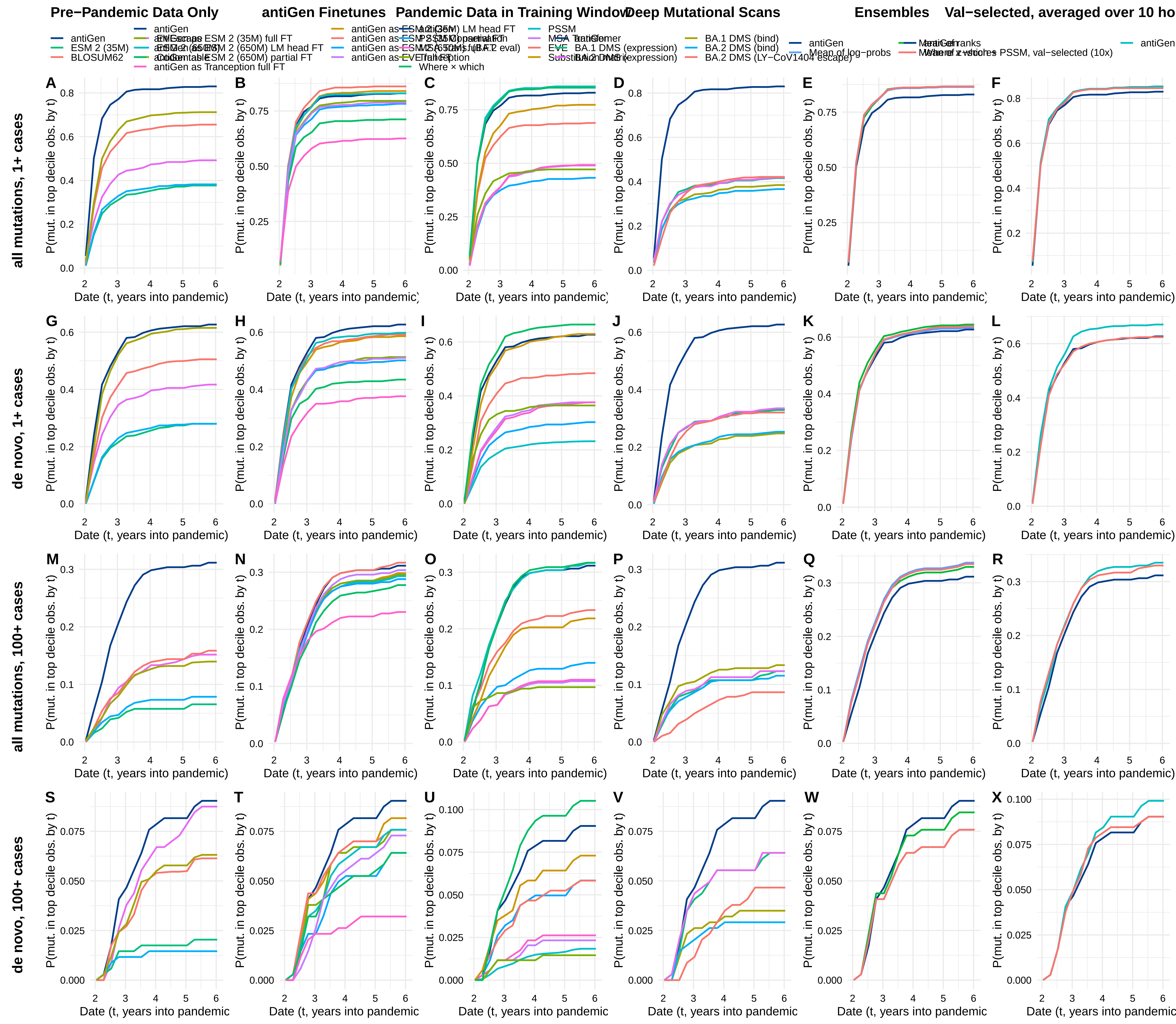
Pre−Pandemic Data Only
antiGen Finetunes
Pandemic Data in Training Window
Deep Mutational Scans
Ensembles
Val−selected, averaged over 10 ho
antiGen
ESM 2 (35M)
BLOSUM62
antiGen
EVEscape
ESM 2 (650M)
Conditionable
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) partial FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as EVE full FT
antiGen
Tranception
Where × which
PSSM
MSA Transformer
EVE
Substitution matrix
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (LY−CoV1404 escape)
antiGen
Mean of log−probs
Mean of ranks
Mean of z−scores
Where × which + PSSM, val−selected (10x)
antiGen
all mutations, 1+ cases
de novo, 1+ cases
all mutations, 100+ cases
de novo, 100+ cases
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
P(mut. in top decile obs. by t)
Date (t, years into pandemic)
0.0
0.1
0.2
0.3
0.4
0.6
0.8
0.00
0.025
0.050
0.075
0.100
0.25
0.50
0.75
2
3
4
5
6